No matter where you look, logs are always in use, so the question is not *if* but *which* backend you should use. We will get back to the selection criteria later in the chapter, but for now, you can rest assured that you will need a log backend. With this, you have a rough idea what logs backends can do and what options exist so let's move on to metrics.

5.3 Backend destinations for metrics

Time series databases (TSDB) are the bread and butter of storing and querying signals of the metrics type. You can use TSDBs to track a range of different metrics, from low-level systems metrics, such as the memory usage of a container, to application-level metrics, such as the number of invocations of your service.

What is a TSDB, and how does it work? As shown in figure 5.2, the main index is the time. The unit of bookkeeping is the metric that comprises one or more time series. Each time series, in turn, keeps track of a value and, typically, you can have many of these time series active at the same time (in the sense that you can use them

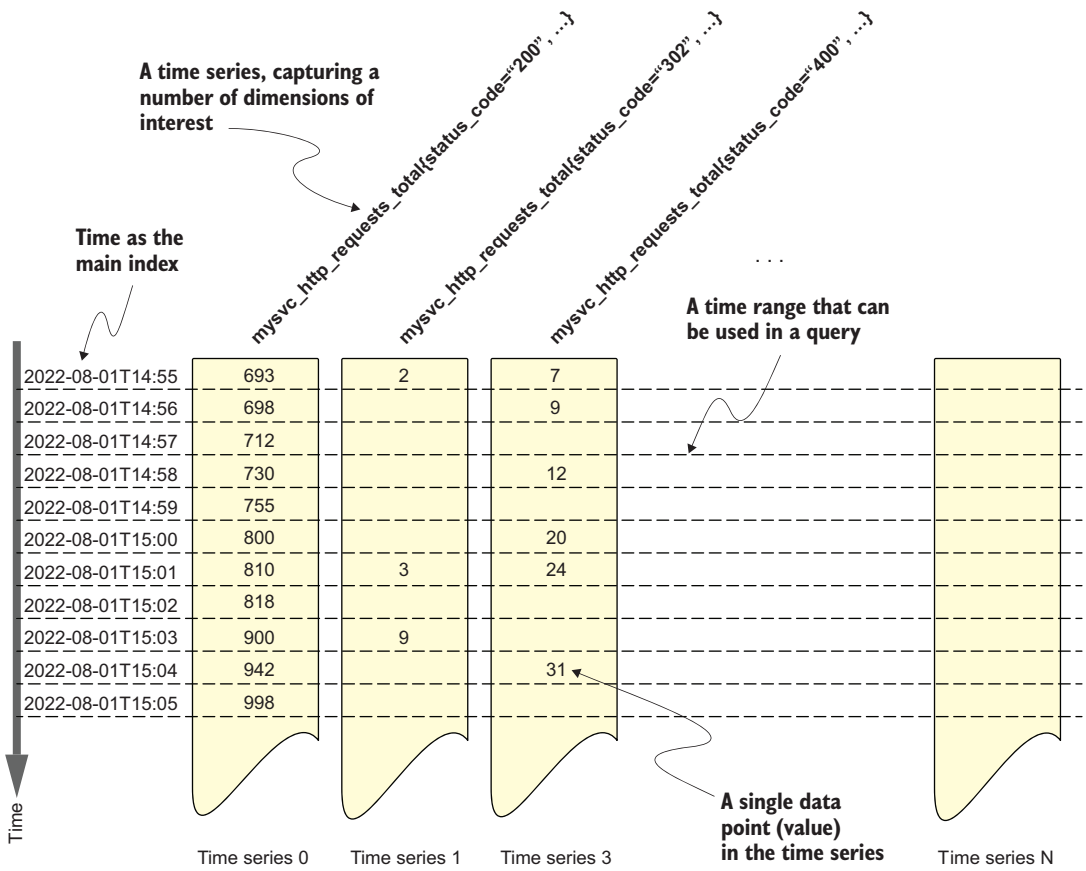


Figure 5.2 Time series database concept, showing N time series of the `mysvc_http_request_total` metric